

Repeated games

1. In an infinitely repeated game, the discount factor δ can be read as ...
 - (a) the probability that the game continues to another round
 - (b) the number of rounds already played
 - (c) the number of players
 - (d) the payoff from defecting
2. The Grudger strategy ...
 - (a) copies the opponent's last move
 - (b) always defects
 - (c) cooperates until the opponent defects, then defects for ever
 - (d) cooperates with probability $1/2$
3. The Folk Theorem says that, for patient enough players, ...
 - (a) cooperation is impossible
 - (b) any individually rational payoff pair can be sustained in a subgame perfect equilibrium
 - (c) the game must be zero sum
 - (d) only the stage Nash equilibrium can occur
4. In the finitely repeated Prisoner's Dilemma with a known last round, ...
 - (a) cooperation is sustained throughout
 - (b) the players cooperate only in the last round
 - (c) there is no subgame perfect equilibrium
 - (d) backward induction forces defection in every round
5. The individually rational payoff of a player is ...
 - (a) the largest payoff in the game
 - (b) the most they can guarantee themselves whatever the opponent does (their minimax value)
 - (c) always zero
 - (d) the payoff from mutual cooperation
6. As the temptation payoff T increases, the discount factor needed to sustain cooperation ...
 - (a) stays the same
 - (b) decreases
 - (c) becomes negative
 - (d) increases (players must be more patient)

7. A strategy in a repeated game is ...
- (a) the payoff matrix
 - (b) the discount factor
 - (c) a single action played every round
 - (d) a mapping from the history of play to an action in the stage game
8. The discounted value of receiving a payoff c every round is ...
- (a) $c\delta$
 - (b) c/δ
 - (c) $c/(1 - \delta)$
 - (d) $c(1 - \delta)$
9. The average utility of a payoff stream is defined as ...
- (a) $(1 - \delta)$ times the discounted total
 - (b) the largest stage payoff
 - (c) the discounted total
 - (d) the payoff in the first round
10. Playing a Nash equilibrium of the stage game in every round is ...
- (a) always a subgame perfect equilibrium of the repeated game
 - (b) impossible to compute
 - (c) never an equilibrium
 - (d) only a Nash equilibrium if the game is finite
11. Cooperation can be sustained in the infinitely repeated Prisoner's Dilemma because ...
- (a) the stage game has no Nash equilibrium
 - (b) players cannot remember the past
 - (c) defection is impossible
 - (d) there is no last round, so future punishment can deter defection
12. The expected number of rounds when δ is the continuation probability is ...
- (a) infinite for any δ
 - (b) $1 - \delta$
 - (c) $1/(1 - \delta)$
 - (d) δ
13. In Grudger, the punishment phase consists of ...

- (a) returning to cooperation after one round
 - (b) alternating cooperation and defection
 - (c) both players playing the stage Nash equilibrium (mutual defection) for ever
 - (d) ending the game immediately
14. For cooperation to be sustainable, the cooperative payoff must exceed ...
- (a) the number of rounds
 - (b) the gain from a one-off defection followed by punishment
 - (c) the discount factor
 - (d) the total of all stage payoffs
15. A larger discount factor δ means ...
- (a) the future matters less
 - (b) the future matters more, making cooperation easier to sustain
 - (c) the game is shorter
 - (d) defection is always best
16. The utility of an infinitely repeated game with discount factor δ is ...
- (a) $\max_i u(s_r(i), s_c(i))$
 - (b) $\sum_{i=0}^{\infty} \delta^i u(s_r(i), s_c(i))$
 - (c) $\sum_{i=0}^{\infty} u(s_r(i), s_c(i))$
 - (d) $\delta u(s_r(0), s_c(0))$
17. An individually rational payoff pair is one that ...
- (a) equals the temptation payoff
 - (b) gives one player everything
 - (c) minimises both players' payoffs
 - (d) exceeds the stage-game Nash equilibrium payoffs for both players
18. For a patient enough population the Folk Theorem constructs a subgame perfect equilibrium achieving ...
- (a) any feasible individually rational payoff pair
 - (b) no payoff at all
 - (c) only the stage Nash payoff
 - (d) only the cooperative payoff
19. In the Folk Theorem's trigger strategy, a deviation is punished by ...
- (a) leaving the game

- (b) forgiving after one round
 - (c) raising the discount factor
 - (d) reverting to the stage-game Nash equilibrium for all future rounds
20. Against a trigger strategy, the most tempting time to deviate is ...
- (a) a randomly chosen stage
 - (b) the first stage, since later gains are discounted more heavily
 - (c) never, deviation always pays
 - (d) the last stage